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BS. Zoology 7th
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Measurement Scales:

Measurement Scales are defined as: The device or an object used to measure or quantify any event or an object is called measurement scales.

There are four types of measurement scales:

Nominal Scale.

Ordinal Scale.

Interval Scale.

Ratio Scale.

Nominal Scale: Nominal

scales are defined as:

"The scales that are used to categorize data into groups."

Numbers serve as tags or labels to classify or identify the objects.

Example Gender, Country, players of any team the T-shirts number's etc.

Ordinal Scale: Ordinal

Scale is defined as:

The scale that are used to measure variables in a natural order, such as rating or ranking.

They provide meaningful insights.

Example: Ranking of class students on the bases of examination (1st, 2nd, 3rd).

Interval Scale: Interval Scale

are defined as:

Scales used to ~~class~~ quantify the data in equal interval between values.

No true absolute zero.

Example: Temperature in Celsius.

IQ levels or scores.

Ratio Scale: Ratio Scale

can be defined as:

Data divided into orders or ranks. Data points can be quantified. Existence of absolute zero values.

Don't have negative values.

Example: Height of a person.
Weight, age, income etc.

Steps involved in research design:

- i) **Identify the research problem's:** Clearly define what you want to study or solve.
- ii) **Review of literature:** Study existing research related to your topic.
- iii) **Formulate objective:** Set clear aims or hypothesis to test.
- iv) **Choose the research design type:** Decide between qualitative quantitative or mixed method.
- v) **Define Variable:** In identifying independent and control variables.

Select the Sample: Choose your target population and sampling method.

Select data Collected method:

Use tools like surveys, interviews, experiments or observations.

Plan for data analysis: Choose statistical tools or quantitative method.

Consider ethical issues: Ensure informed consent, confidentiality and ethical approve.

Prepare research timeline: Set deadlines and estimate costs.

Finalize research design: Review everything and get ready for implementation.

Dependent Variables: The

variables that you measure or observe its effects or outcomes.

Example: Growth of plants, number of leaves.

Independent Variables: The

variables that you ^{can} change or control in an experiment.

Example: Type of Fertilizers